

QUESTION 1

SCEGGS REDLANDS 1998 3U

(a) (i) Find $\int \frac{dx}{x^2 + 4}$

(ii) Find $\int \frac{x^2 dx}{x^3 - 8}$

(b) Evaluate: (i) $\int_2^7 \frac{x dx}{\sqrt{x+2}}$ using the substitution $u = x+2$

(ii) $\int_2^7 \frac{x dx}{\sqrt{x+2}}$ using the substitution $u = \sqrt{x+2}$

(c) (i) Show that $\tan x \equiv \frac{\sin 2x}{1 + \cos 2x}$

(ii) Hence evaluate $\tan \frac{\pi}{12}$.

QUESTION 2:

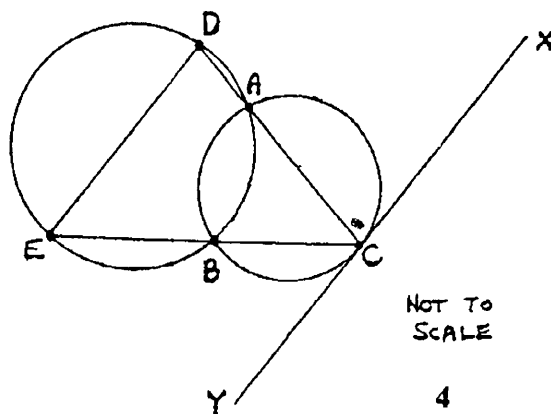
- (a) A is the point (3,2) and B is the point (x_B, y_B) .
The point P (-4,2) divides AB internally in the ratio 2 : 5
(i.e. AP : PB = 2 : 5). Find the values of x_B and y_B .

(b) (i) If $f(x) = \sin^{-1} \frac{x}{2}$ find $f^{-1}(x)$.

(ii) State the domain and range of $f^{-1}(x)$.

(iii) Sketch the graph of $3y = \sin^{-1} \frac{x}{2}$ stating clearly its domain and range.

- (c) Two circles intersect at A and B.
From any point C on the smaller circle lines CAD and CBE are drawn cutting the larger circle at D and E respectively.
XY is the tangent at C.
Prove formally that DE is parallel to XY.



QUESTION 3:

12 marks (Start a new answer booklet)

MARKS

- (a) In the diagram P and Q are two points on the parabola $x^2 = 4ay$ having coordinates respectively of $(2ap, ap^2)$ and $(2aq, aq^2)$.

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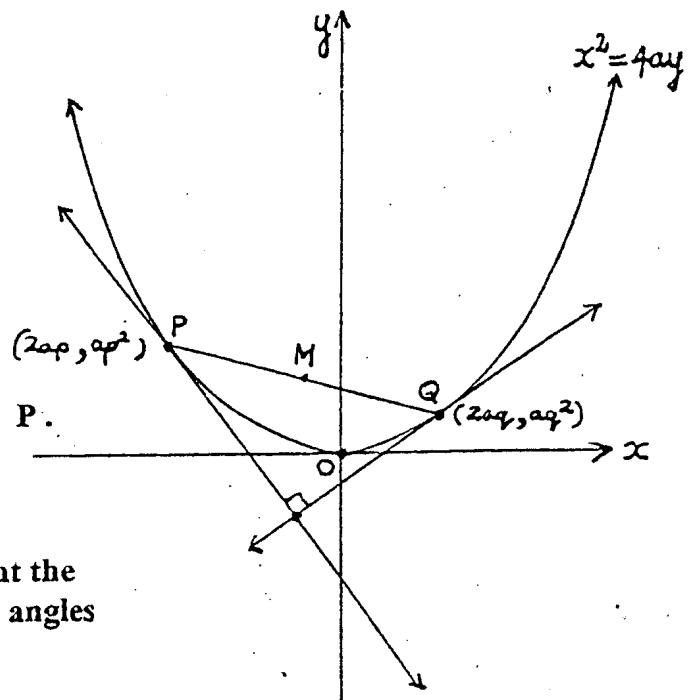
- (i) Show that the equation of PQ is $y = \left(\frac{p+q}{2}\right)x - apq$.

- (ii) Find the gradient of the tangent at P.

- (iii) Hence write down the condition that the tangents at P and Q are at right angles to each other.

- (iv) What are the coordinates of the midpoint M of PQ?

- (v) Show that the locus of M, as the points P and Q move around the parabola with the tangents at P and Q being perpendicular to each other, is another parabola with equation $x^2 = 2a(y-a)$. Write down the coordinates of the vertex and focus of this locus parabola.



NOT TO SCALE

- (b) Air is being pumped into a spherical balloon at the rate of $450 \text{ cm}^3/\text{sec}$.

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Given that the volume of a sphere is given by $V = \frac{4}{3}\pi r^3$, calculate the rate at which the radius of the balloon is increasing at the instant when its radius reaches 15 cm.

QUESTION 4: 12 marks (Start a new answer booklet)

MARKS

- (a) If $f(x) = x^3 + 3x^2 - 10x - 24$ 3

Calculate $f(-2)$ and express $f(x)$ as the product of three linear factors.

- (b) If α, β, γ are the roots of the equation $x^3 + 2x^2 - 3x + 5 = 0$ 3
state the values of :

(i) $\alpha + \beta + \gamma$

(ii) $\alpha\beta + \beta\gamma + \gamma\alpha$

(iii) $\alpha\beta\gamma$

(iv) *Hence calculate the value of $(\alpha - 1)(\beta - 1)(\gamma - 1)$.*

- (c) Solve for x given that $\frac{2x+3}{x-4} > 1$. 3
Sketch your solution on a number line.

- (d) *Differentiate with respect to x :* 3

(i) $y = x \sin^{-1} \frac{x}{2}$

(ii) $y = \tan(x^3)$

(iii) $y = \frac{e^{2x}}{1 + \cos x}$

QUESTION 5:

12 marks (Start a new answer booklet)

MARKS

(a) (i) Show that $\frac{\sin \alpha + \sin \beta}{\sin \alpha - \sin \beta} = \frac{\tan \frac{\alpha + \beta}{2}}{\tan \frac{\alpha - \beta}{2}}$.

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(ii) Find the most general solution for θ satisfying the equation $4 \sin^2 \theta - 1 = 0$.

- (b) A body is heated to a temperature of 120°C and left to cool in a room whose room temperature is 20°C . After 10 minutes the temperature of the body cools to 80°C .

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You may assume that the rate of cooling can be expressed in the differential equation

$$\frac{dT}{dt} = -k(T - 20) \quad .$$

- (i) Show by integration that the temperature T can be expressed in the form

$$T = 20 + 100e^{-kt} \quad \text{where} \quad k = -\frac{1}{10} \ln \frac{3}{5} \quad .$$

- (ii) What will be the temperature to the nearest degree of the body after a further 25 minutes?

QUESTION 6:

12 marks

(Start a new answer booklet)

MARI

- (a) The speed $|v|$ of a particle moving along the x -axis is given by the equation

$$v^2 = 12 + 8x - 4x^2$$

where x is the displacement of the point from the origin.

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- (i) *Prove* that the motion is simple harmonic.
- (ii) *Find* its centre of motion.
- (iii) *Calculate* its period.
- (iv) *Show* that its amplitude is 2 units.

- (b) (i) *Write down* an expression for $\sin^2 \theta$ in terms of $\cos 2\theta$.

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- (ii) *Hence evaluate* $\int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta$.

- (c) (i) *Sketch* the curve $y = 1 + \sin x$ for the domain $-\pi \leq x \leq \pi$.

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- (ii) *Hence sketch* the shape of the solid of revolution formed by rotation of this curve about the x -axis .

- (iii) *Show* that the volume of this solid formed by rotation about the x -axis is $3\pi^2$ units².

QUESTION 7: 12 marks (Start a new answer booklet)

MARKS

- (a) A projectile P is projected with initial velocity U at angle α to the horizontal. 8

Show by using $\ddot{x} = 0$ and $\ddot{y} = -g$ and without assuming a numerical value for g that :

- (i) The time taken to reach maximum height is given by

$$t = \frac{U \sin \alpha}{g} .$$

- (ii) Find this maximum height reached by the projectile.

- (iii) Show that to obtain a maximum range, the angle of projection must be 45° .

- (b) A missile is projected with a speed of 100 m/s at an elevation of 45° aimed at a tall building which is a horizontal distance of 400 m from the point of projection. 4

- (i) Find the time of flight until the missile strikes the building.

- (ii) Find how high on the building the missile strikes. (You may use the approximation $g \approx 10 \text{ m/s}^2$ for this part, ie part (ii) .)

Question NO. 1

(i) $\int \frac{dx}{x^2+4} = \frac{1}{2} \tan^{-1} \frac{x}{2} + C$ (1)

(ii) $\int \frac{x^2 dx}{x^3-8} = \frac{1}{3} \log_e(x^3-8) + C$ (1)

(b) (i) $I = \int_2^7 \frac{x dx}{\sqrt{x+2}}$ $u = x+2, x = u-2$
 $\therefore du = dx$
 When $x=2, u=4$
 When $x=7, u=9$

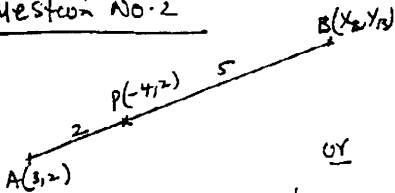
$\therefore I = \int_4^9 \frac{(u-2) du}{u^{1/2}}$ (3)
 $= \int_4^9 (u^{1/2} - 2u^{-1/2}) du$
 $= \left[\frac{2}{3} u^{3/2} - 4u^{1/2} \right]_4^9$
 $= \left(\frac{2}{3} \times 27 - 4 \times 3 \right) - \left(\frac{2}{3} \times 8 - 4 \times 2 \right)$
 $= 18 - 12 - 5\frac{1}{3} + 8$
 $= 8\frac{2}{3} \text{ Units.}$

(ii) $I = \int_2^7 \frac{x dx}{\sqrt{x+2}}$ $u = \sqrt{x+2}$
 $\therefore u^2 = x+2$
 $\therefore 2u du = dx$
 When $x=2, u=2$
 When $x=7, u=3$

$= \int_2^3 \frac{(u^2-2) \cdot 2u du}{u}$ (2)
 $= 2 \int_2^3 (u^2-2) du$
 $= 2 \left[\frac{u^3}{3} - 2u \right]_2^3$
 $= 2 \left[\left(\frac{27}{3} - 6 \right) - \left(\frac{8}{3} - 4 \right) \right]$
 $= 8\frac{2}{3} \text{ Units.}$

Question NO. 2

(a).

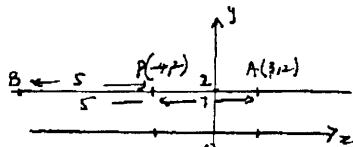


$\frac{2x_B + 15}{7} = -4$
 $\therefore 2x_B + 15 = -28$
 $\therefore 2x_B = -43$
 $\therefore x_B = -21\frac{1}{2}$ (1)

$\frac{2y_B + 10}{7} = 2$
 $\therefore 2y_B + 10 = 14$
 $\therefore 2y_B = 4$
 $\therefore y_B = 2$ (1)

$\therefore B$ is the point $(-21\frac{1}{2}, 2)$

OR



AP = 7 units which is equivalent to 1 part = $3\frac{1}{2}$ units.

PB corresponds to 5 parts
 $\therefore$ length PB = $5 \times 3\frac{1}{2}$ units
 $= 17\frac{1}{2}$ units

$\therefore$ Co-ords of P are $(-21\frac{1}{2}, 2)$

(b) (i) $f(x) = \sin^{-1} \frac{x}{2}$ (2)
 $f: y = \sin^{-1} \frac{x}{2}$
 $f^{-1}: x = \sin \frac{y}{2}$

$\therefore \sin x = \frac{y}{2}$
 $\therefore y = 2 \sin x$

D: $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$
 R: $-1 \leq \frac{y}{2} \leq 1$ (1)

c). To show $\tan x = \frac{\sin 2x}{1 + \cos 2x}$

RHS = $\frac{\sin 2x}{1 + \cos 2x}$ (2)
 $= \frac{2 \sin x \cos x}{1 + 2 \cos^2 x - 1}$
 $= \frac{2 \sin x \cos x}{2 \cos^2 x}$
 $= \frac{\sin x}{\cos x} \cdot \frac{\cos x}{\cos x}$
 $= \tan x$
 $= \text{LHS}$

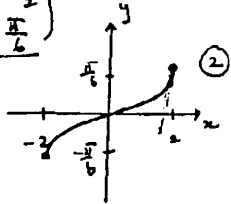
Using the above: $\tan \frac{\pi}{12} = \frac{\sin 2 \times \frac{\pi}{12}}{1 + \cos 2 \times \frac{\pi}{12}}$ (2)
 (Replace x by $\frac{\pi}{12}$)
 $= \frac{\sin \frac{\pi}{6}}{1 + \cos \frac{\pi}{6}}$
 $= \frac{\frac{1}{2}}{1 + \frac{\sqrt{3}}{2}}$
 $= \frac{1}{2 + \sqrt{3}}$
 (or $= 2 - \sqrt{3}$)

Total 12 Marks

(iii) $3y = \sin^{-1} \frac{x}{2}$

D: $-1 \leq \frac{x}{2} \leq 1$
 ie $-2 \leq x \leq 2$ (1)

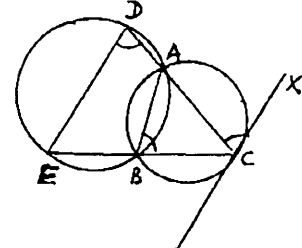
R: $-\frac{\pi}{2} \leq 3y \leq \frac{\pi}{2}$
 ie $-\frac{\pi}{6} \leq y \leq \frac{\pi}{6}$



(c)

(4)

Data: XY is a tangent at C
 CA, CB, CE and ED are radii.
 To Prove: DE || XY
 Constr: Join AB



Proof:
 $\angle ACX = \angle ABC$ — Angles in the same segment — XY is a tangent
 $\angle ABC = \angle ADE$ — Exterior angle of cyclic quad ABED
 $\therefore \angle ACX = \angle ADE$
 But these are alternate angles to lines XY and DE
 $\therefore XY \parallel DE$

Total 12 Marks

Question 3 (continued)

iii) The tangents are lt if

$$m_1 \times m_2 = -1$$

$$\therefore p \times q = -1 \Rightarrow pq = -1$$

$$\text{iv) } M \left(a(p+q), \frac{a(p^2+q^2)}{2} \right)$$

$$x_M = a(p+q)$$

$$y_M = \frac{a}{2}(p^2+q^2)$$

$$\text{and } pq = -1$$

$$\begin{aligned} \therefore y &= \frac{a}{2} \left[(p+q)^2 - 2pq \right] \\ &= \frac{a}{2} \left[\frac{x^2}{a^2} - 2pq \right] \end{aligned}$$

$$y = \frac{x^2}{2a} - \frac{a}{2} \times 2 \times -1$$

$$y = \frac{x^2}{2a} + a$$

$$\therefore x^2 = 2a(y-a)$$

$$x^2 = 4 \times \frac{a}{2} (y-a)$$

$$\therefore \text{Vertex at } (0, a)$$

$$\text{Focus at } \left(0, \frac{3a}{2}\right)$$

$$\text{b) } \frac{dv}{dt} = 450 \text{ cm}^3/\text{sec.}$$

$$V = \frac{4}{3} \pi R^3$$

$$\therefore \frac{dv}{dR} = 4\pi R^2$$

$$\therefore \frac{dR}{dt} = \frac{dR}{dv} \cdot \frac{dv}{dt}$$

$$\therefore \frac{dR}{dt} = \frac{1}{4\pi \times 15^2} \times 450 \quad (R = 15)$$

$$\frac{dR}{dt} = \frac{1}{2\pi} \text{ cm/sec.}$$

$$\approx 0.16 \text{ cm/sec.}$$

Question NO. 4

(a) $f(x) = x^3 + 3x^2 - 10x - 24$

$$f(-2) = (-2)^3 + 3(-2)^2 - 10(-2) - 24$$

$$= -8 + 12 + 20 - 24$$

$$= 0$$

$\therefore (x+2)$ is a factor

$$\begin{array}{r} x^2 + x - 12 \\ x+2 \overline{) x^3 + 3x^2 - 10x - 24} \\ \underline{x^2 + 2x^2} \\ x^2 + 2x^2 \end{array}$$

$$\therefore f(x) = (x+2)(x^2 + x - 12)$$

$$= (x+2)(x+4)(x-3)$$

$$\begin{array}{r} x^2 - 10x \\ x^2 + 2x \\ \hline -12x - 24 \\ -12x - 24 \end{array}$$

(b) $x^3 + 2x^2 - 3x + 5 = 0$ (α, β, γ)

(i) $\alpha + \beta + \gamma = -2$

(ii) $\alpha\beta + \beta\gamma + \alpha\gamma = -3$

(iii) $\alpha\beta\gamma = -5$

(iv) $(\alpha-1)(\beta-1)(\gamma-1)$

$$= (\alpha-1)(\beta\gamma - \beta - \gamma + 1)$$

$$= \alpha\beta\gamma - \alpha\beta - \alpha\gamma + \alpha - \beta\gamma + \beta + \gamma - 1$$

$$= \alpha\beta\gamma - (\alpha\beta + \beta\gamma + \alpha\gamma) + (\alpha + \beta + \gamma) - 1$$

$$= -5 - (-3) + (-2) - 1$$

$$= -5 + 3 - 2 - 1 = -5$$

Question NO. 2

(a) (i) Show that $\frac{\sin \alpha + \sin \beta}{\sin \alpha - \sin \beta} = \frac{\tan \frac{\alpha+\beta}{2}}{\tan \frac{\alpha-\beta}{2}}$

$$\text{LHS} = \frac{\sin \alpha + \sin \beta}{\sin \alpha - \sin \beta}$$

$$= \frac{2 \sin \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}}{2 \cos \frac{\alpha+\beta}{2} \sin \frac{\alpha-\beta}{2}}$$

$$= \frac{\tan \frac{\alpha+\beta}{2} \cdot \cot \frac{\alpha-\beta}{2}}{\tan \frac{\alpha-\beta}{2}}$$

(ii) $4 \sin^2 \theta = 1$

$$\therefore \sin^2 \theta = \frac{1}{4}$$

$$\therefore \sin \theta = \pm \frac{1}{2}$$

$$\therefore \theta = n\pi \pm \frac{\pi}{6}$$

(b) (i) $\frac{dT}{dt} = -k(T-20)$

$$\therefore \frac{dT}{T-20} = -k dt$$

$$\therefore \int \frac{dT}{T-20} = -k \int dt + A$$

$$\therefore \log_e(T-20) = -kt + A$$

$$T-20 = e^{-kt+A} = e^{-kt} \cdot e^A$$

(c) $\frac{2x+3}{x-4} > 1$ $x \neq 4$

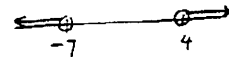
$$\therefore \frac{2x+3}{x-4} \cdot (x-4)^2 > (x-4)^2$$

$$\text{i.e. } (2x+3)(x-4) > (x-4)^2$$

$$\text{i.e. } (x-4)[2x+3 - (x-4)] > 0$$

$$\text{i.e. } (x-4)(x+7) > 0$$

$$\text{i.e. } x > 4 \text{ or } x < -7$$



(d) (i) $y = x \sin^{-1}\left(\frac{x}{2}\right)$

$$\frac{dy}{dx} = \sin^{-1}\left(\frac{x}{2}\right) \cdot 1 + x \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{1-\left(\frac{x}{2}\right)^2}}$$

$$= \sin^{-1}\frac{x}{2} + \frac{x}{\sqrt{4-x^2}}$$

(ii) $y = \tan(x^3)$

$$\frac{dy}{dx} = 3x^2 \cdot \sec^2(x^3)$$

(iii) $y = \frac{e^{2x}}{1+\cos x}$

$$\frac{dy}{dx} = \frac{(1+\cos x) \cdot 2e^{2x} - e^{2x}(0-\sin x)}{(1+\cos x)^2}$$

$$= \frac{e^{2x}(2+2\cos x + \sin x)}{(1+\cos x)^2}$$

Total: 12 marks

$$T = 20 + Be^{-kt} \quad \text{--- (1)}$$

When $t=0$, $T = 120^\circ\text{C}$

$$\therefore 120 = 20 + B$$

$$\therefore B = 100$$

$$\therefore T = 20 + 100e^{-kt}$$

When $t = 10 \text{ min}$, $T = 80^\circ\text{C}$

$$\therefore \text{Subm (1):}$$

$$80 = 20 + 100e^{-10k}$$

$$\therefore 60 = 100e^{-10k}$$

$$\therefore e^{-10k} = \frac{60}{100} = \frac{3}{5}$$

$$\therefore -10k = \log_e \frac{3}{5}$$

$$\therefore k = -\frac{1}{10} \log_e \frac{3}{5}$$

$$\therefore T = 20 + 100e^{-\frac{1}{10}(\log_e \frac{3}{5})t}$$

$$\therefore T = 20 + 100e^{-\frac{1}{10}(\log_e \frac{3}{5})t}$$

After a further 25 mts:
total time taken to cool will be 35 mts.

$$\therefore T = 20 + 100e^{-\frac{1}{10}(\log_e \frac{3}{5})35}$$

$$= 36.73^\circ (37^\circ)$$

Total: 12 marks

Question 6

a) i) $v^2 = 12 + 8x - 4x^2$

$$\frac{v^2}{2} = 6 + 4x - 2x^2$$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 4 - 4x$$

$$\ddot{x} = -4(x-1)$$

This is in the form $\ddot{x} = -4x$
($X = x-1$)

$\therefore$ The motion is S.H.M.

ii) Centre of motion $x = 1$

iii) $T = \frac{2\pi}{n} = \frac{2\pi}{2}$

$$T = \pi \text{ sec.}$$

iv) max. displacement when $v = 0$,

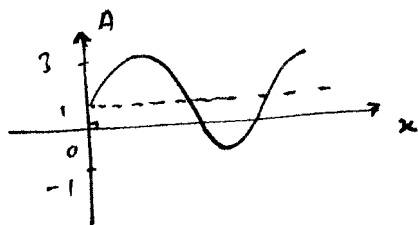
$$\therefore 12 + 8x - 4x^2 = 0$$

$$\therefore 4(-x^2 + 2x + 3) = 0$$

$$\therefore x^2 - 2x - 3 = 0$$

$$(x-3)(x+1) = 0$$

$$\therefore x = -1, x = 3$$



Amplitude is $A = 2$

b) i) $\cos 2\theta = 1 - 2\sin^2 \theta$

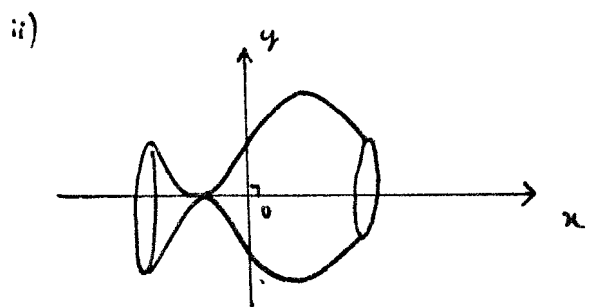
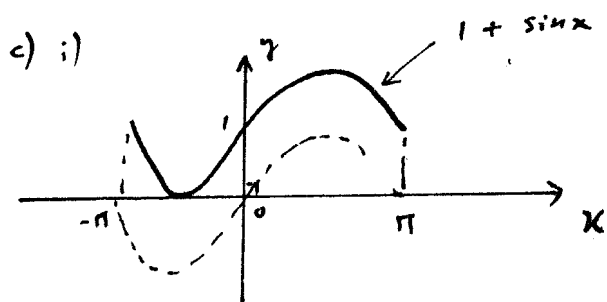
$$\therefore 2\sin^2 \theta = 1 - \cos 2\theta$$

$$\therefore \sin^2 \theta = \frac{1}{2} (1 - \cos 2\theta)$$

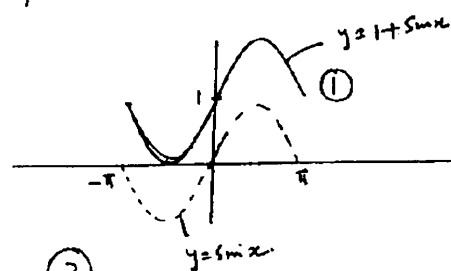
ii)
$$\int_0^{\pi/2} \sin^2 \theta d\theta = \frac{1}{2} \int_0^{\pi/2} (1 - \cos 2\theta) d\theta$$

$$= \frac{1}{2} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\pi/2}$$

$$= \pi/4$$



c) $y = 1 + \sin x$ for $-\pi \leq x \leq \pi$



iii)

$$V = \int_{-\pi}^{\pi} y^2 dx$$

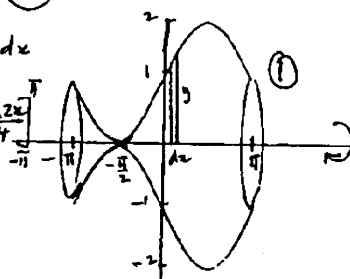
$$= \int_{-\pi}^{\pi} (1 + \sin x)^2 dx \quad (2)$$

$$= \int_{-\pi}^{\pi} (1 + 2\sin x + \sin^2 x) dx$$

$$= \int_{-\pi}^{\pi} \left[x - 2\cos x + \frac{x}{2} - \frac{\sin 2x}{4} \right] dx$$

$$= \left[\left(\frac{3\pi}{2} + 2 \right) - \left(-\frac{3\pi}{2} + 2 \right) \right]$$

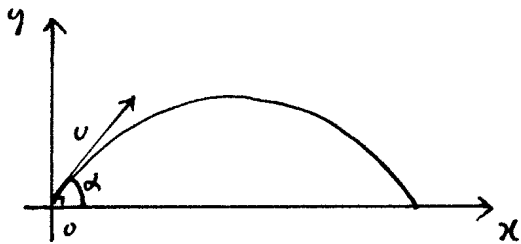
$$= 3\pi^2 \text{ units}^3$$



Total: 12 marks

Question 7

a)



Vertical component

$$\ddot{y} = -g \Rightarrow \dot{y} = -gt + C_1$$

$$t = 0 \quad \dot{y} = U \sin \alpha$$

$$\therefore \dot{y} = -gt + U \sin \alpha$$

$$\therefore y = -\frac{1}{2}gt^2 + (U \sin \alpha)t + C_2$$

$$t = 0, y = 0 \Rightarrow C_2 = 0$$

$$\therefore y = -\frac{1}{2}gt^2 + (U \sin \alpha)t$$

Time to reach max. height as

$$\dot{y} = 0$$

$$\Rightarrow -gt + U \sin \alpha \Rightarrow t = \frac{U \sin \alpha}{g}$$

$$\therefore y_{\max} = -\frac{1}{2}g \times \frac{(U \sin \alpha)^2}{g^2} +$$

$$U \sin \alpha \times \frac{U \sin \alpha}{g}$$

$$y_{\max} = \frac{1}{2g} (U \sin \alpha)^2$$

Horizontal Component

$$\ddot{x} = 0$$

$$\therefore \dot{x} = C_3 = U \cos \alpha$$

$$\therefore x = U \cos \alpha t$$

Time to complete the range

$$T = \frac{2U \sin \alpha}{g}$$

$$\therefore \text{Range} = U \cos \alpha \cdot T$$

$$= U \cos \alpha \times \frac{2U \sin \alpha}{g}$$

$$\text{Range} = \frac{U^2 \sin 2\alpha}{g}$$

$$\text{Range Max if } 2\alpha = 90^\circ$$

$$\therefore \alpha = 45^\circ$$

b) $x = U \cos \alpha t$

$$\therefore 400 = 100 \cos 45^\circ t$$

$$\therefore t = \frac{400}{100 \cos 45^\circ} = 4\sqrt{2} \text{ sec}$$

$$y = -5t^2 + 100 \sin 45^\circ t$$

$$t = 4\sqrt{2} \text{ sec.}$$

$$\therefore y = -5 \times (4\sqrt{2})^2 + 50\sqrt{2} \times 4\sqrt{2}$$

$$y = 240 \text{ m (the height of the building)}$$

